

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating only

\$59.76

Avg. Purchase

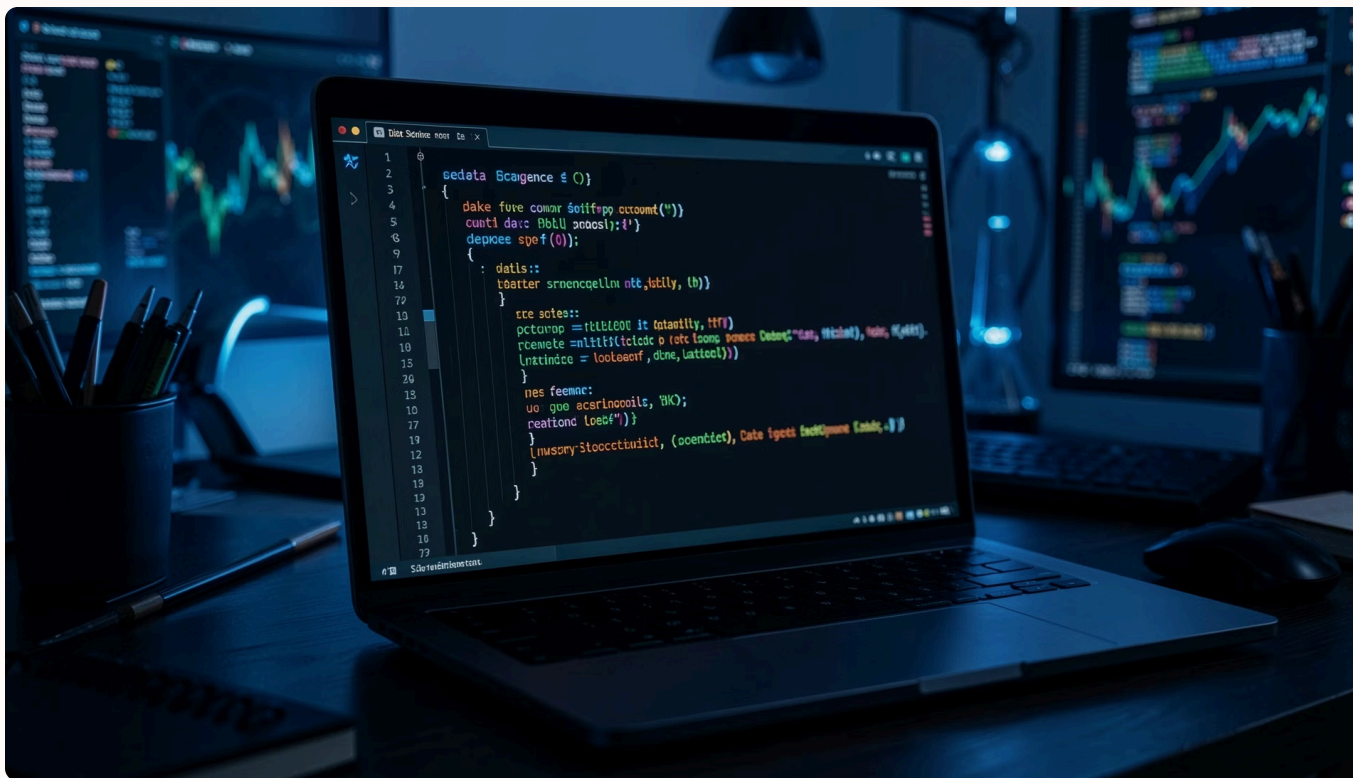
Customer Data

Age, Gender, Location, Subscription Status

Purchase Data

Item, Category, Amount, Season, Discount, Shipping

Python: Data Preparation



01

Load & Explore

pandas, `df.info()`, `.describe()`

02

Clean Missing Data

Median imputation by category

03

Feature Engineering

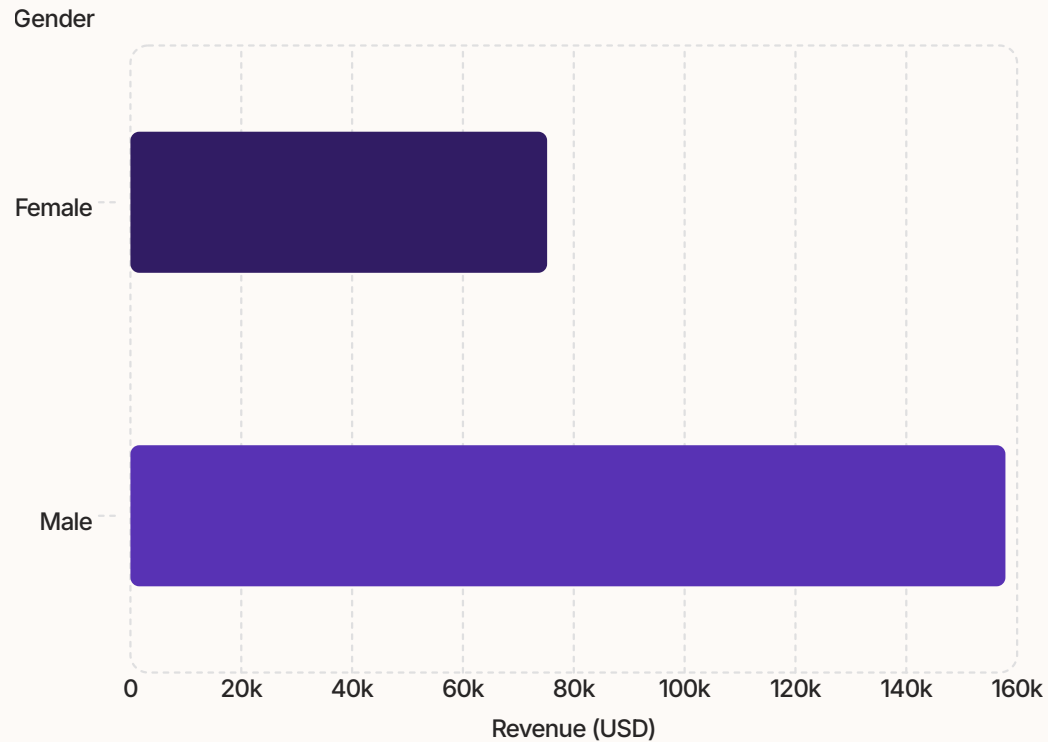
`age_group`, `purchase_frequency_days`

04

Load to PostgreSQL

Ready for SQL analysis

Revenue by Gender



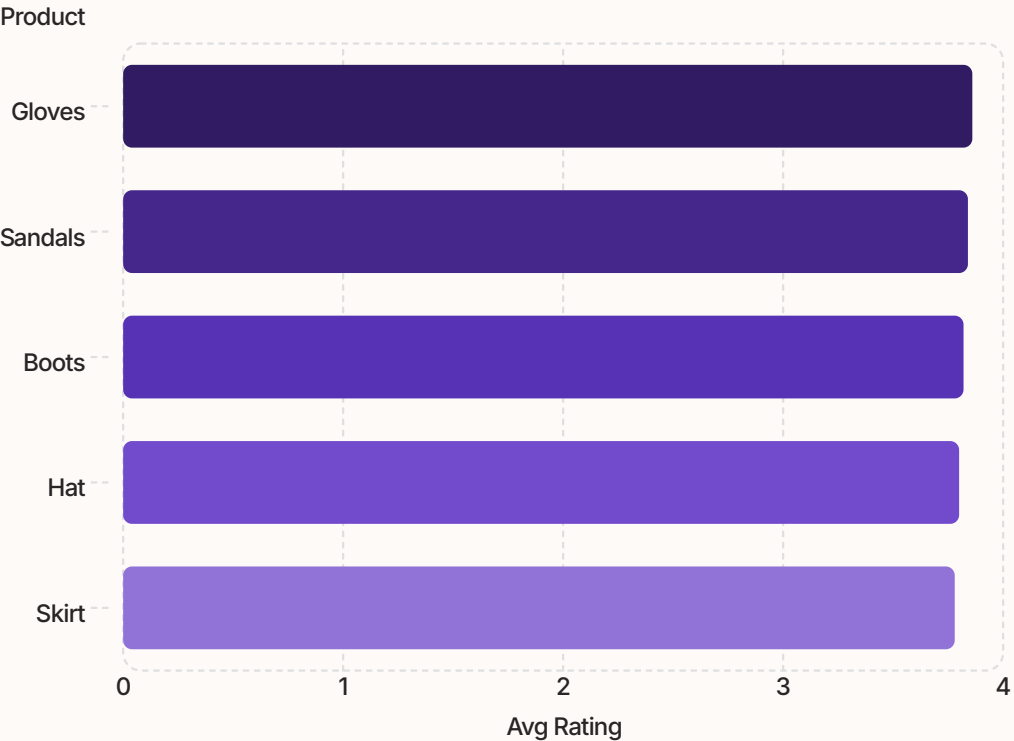
Key Finding

Male customers generate **2.1x more revenue** than female customers.

Male: \$157,890 · Female: \$75,191

Top Products & Shipping Insights

Top 5 Products by Rating



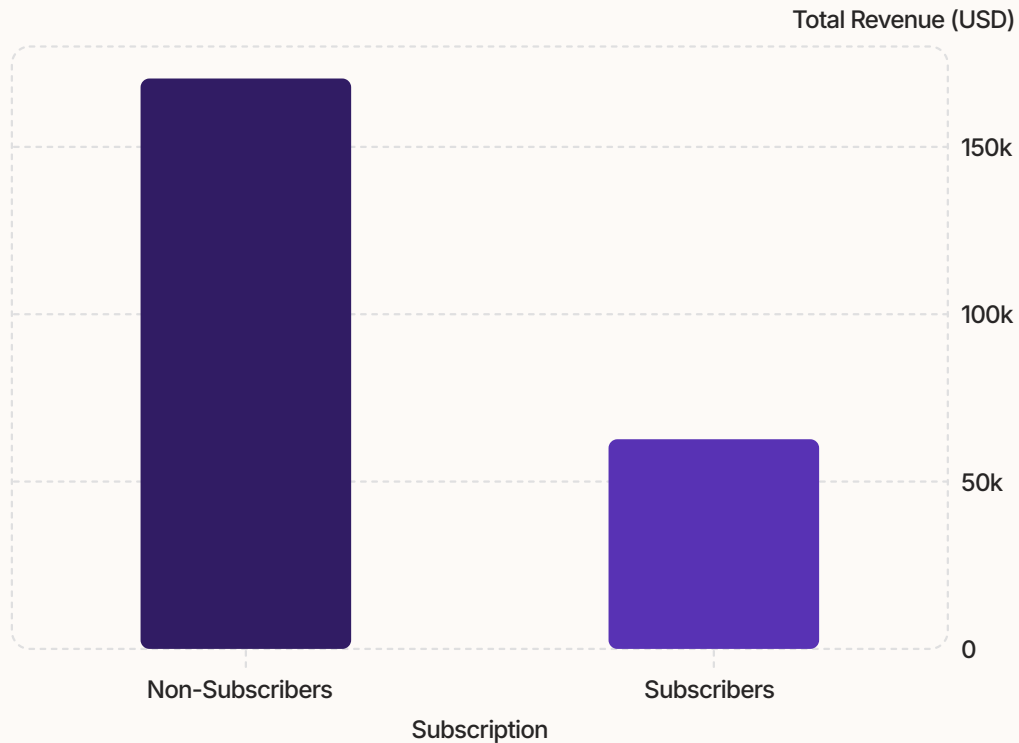
Shipping Type Comparison

Express: **\$60.48** avg · Standard: **\$58.46** avg

Discount-Dependent Products

Hat (50%), Sneakers (49.7%), Coat (49.1%) most discounted

Subscribers vs. Non-Subscribers



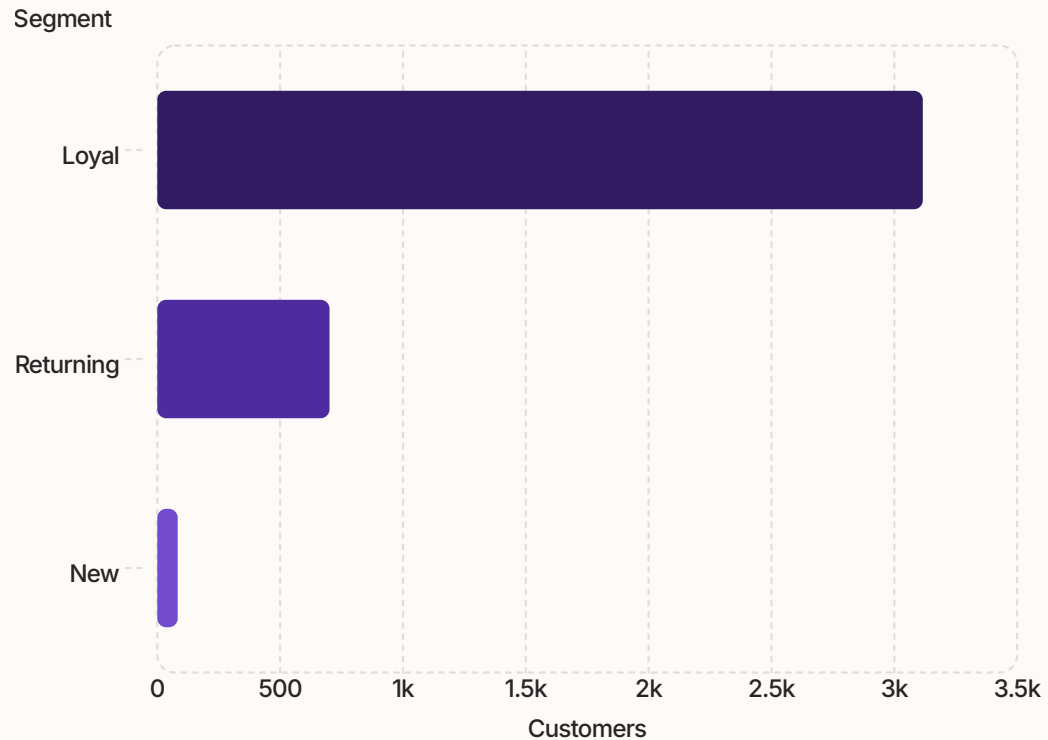
Key Findings

Non-subscribers drive **73%** of total revenue

Avg spend nearly identical: \$59.87 vs \$59.49

958 repeat buyers (5+ purchases) are subscribers

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

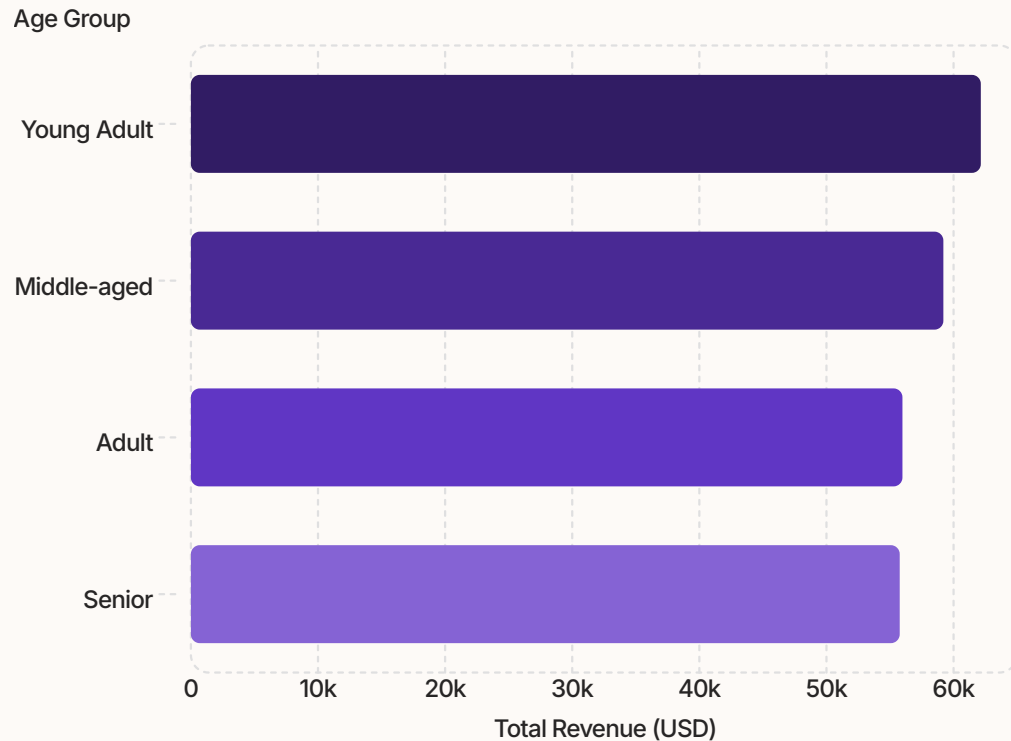
Returning — 18%

701 customers

New — 2%

83 customers

Revenue by Age Group



Insight

Young Adults lead revenue, but all groups are **relatively balanced**

Age range: 18–70 · Avg: 44 years

Power BI Dashboard

Key Metrics

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription · Gender ·
Category · Shipping Type

Visuals

Revenue by category & age group · Sales charts · Subscription donut



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales boost with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users